AI Vision Extraction App

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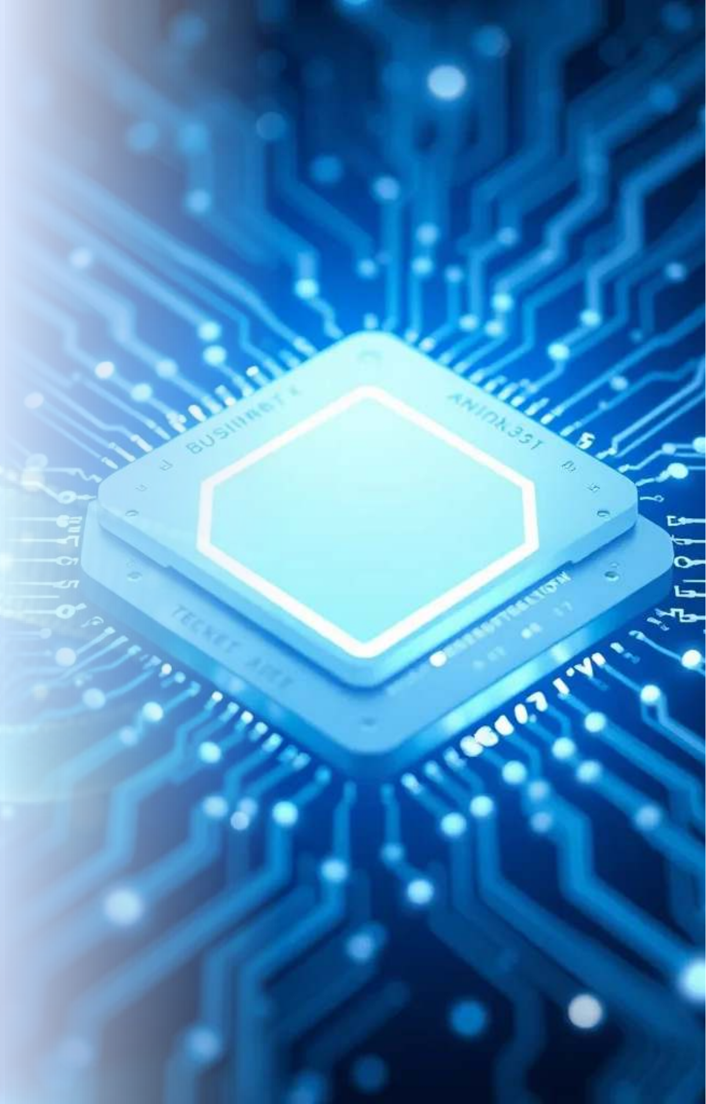
1. Introduction

Object extraction is a crucial step in image processing and computer vision.

AI can automatically detect and isolate objects from images, reducing manual effort.

This project uses DeepLabV3+ segmentation model trained on COCO dataset to extract objects efficiently.



2. Scope

Automate object extraction from images for various industries.

Trained on COCO dataset, which contains diverse objects and complex backgrounds.

Provides a fast, accurate, and interactive tool for users.

3. Importance

1

Saves time and effort by replacing manual segmentation.

2

Ensures high-quality results even for complex images.

3

Supports research, commercial, and educational applications.

4

Can enhance image editing, data labeling, and analysis tasks.

4. Existing System

Traditional systems rely on manual annotation or basic thresholding methods.

Accuracy depends heavily on human effort.

Existing AI-based systems often produce noisy masks or slow predictions.

Limited interactivity in real-time applications.

5. Proposed System

Uses DeepLabV3+, trained on COCO dataset, for automated object extraction.

Implements Test Time Augmentation (TTA) for improved predictions.

Applies post-processing (thresholding + morphology) for cleaner masks.

Interactive web application built with Streamlit for instant preview & download.

6. Dataset

COCO Dataset: Large-scale dataset for object detection, segmentation, and captioning.

Contains over 200,000 labeled images with multiple object categories.

Provides diverse images ensuring model generalizes well to real-world scenarios.

7. Metrics

Metric	Value
Pixel Accuracy	84%
Intersection over Union (IoU)	54%

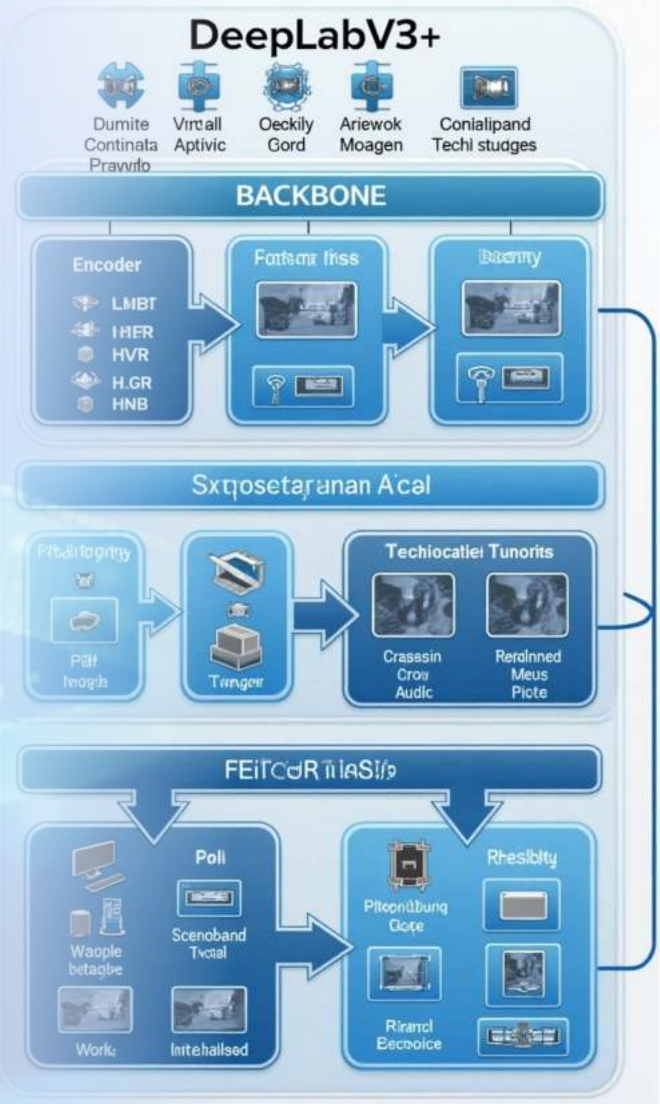
TTA improves accuracy → stabilizes predictions and reduces errors. Post-processing (threshold + morphology) → produces cleaner masks.

8. Architecture

Backbone: DeepLabV3+ encoder (supports multiple encoder options).

Segmentation Head: Atrous Spatial Pyramid Pooling (ASPP) for multi-scale features.

Enhanced Decoder: Refines object boundaries for accurate masks.



9. Pipeline

1. Image Upload → User uploads an image (JPG, JPEG, or PNG).
2. Preprocessing → Resize, normalize, convert to tensor.
3. Segmentation Model → DeepLabV3+ predicts the object mask.
4. TTA → Apply multiple augmentations and average predictions.
5. Post-processing → Threshold + Morphological operations for clean mask.
6. Object Extraction → Overlay mask on original image, black background for removed areas.
7. Download Result → Extracted object available as PNG.\

(Add pipeline diagram or flowchart here)

10. Applications

E-commerce

Automatic product image background removal.

Medical Imaging

Isolate organs or tumors for analysis.

Autonomous Vehicles

Detect and track objects in real-time.

Image Editing & AR

Enhance creativity by isolating objects.

Data Labeling

Speed up dataset annotation for AI models.

11. Conclusion

AI Vision Extraction App provides fast, accurate, and interactive object extraction.

Trained on COCO dataset, ensuring robust generalization.

Incorporates TTA and post-processing for improved mask quality. Can be applied across multiple industries and domains.

Thank You